

Lab Report

Static Routing Configuration Using Cisco Packet Tracer

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1. Objective

- To understand the concept of routing in computer networks.
- To configure static routing between two LANs.
- To establish communication between different networks.
- To configure routers, switches, and PCs in Cisco Packet Tracer.
- To test connectivity using ping command.

2. Software Required

- Cisco Packet Tracer

3. Theory

3.1 Routing

Routing is the process of selecting a path for network traffic in order to send data packets from source to destination across interconnected networks. Routers perform routing by using routing tables and forwarding packets to the appropriate network.

A router is a networking device that connects multiple networks and determines the best path for data transmission.

3.2 Types of Routing

Routing can mainly be classified into the following types:

3.2.1 1. Static Routing

Static routing is a routing method in which routes are manually configured by the network administrator. The router does not automatically update routing information.

Advantages:

- Simple and secure
- Less bandwidth consumption
- Suitable for small networks

Disadvantages:

- Manual configuration required
- Difficult to manage in large networks

3.2.2 2. Dynamic Routing

Dynamic routing automatically updates routing tables using routing protocols such as RIP, OSPF, and EIGRP.

Advantages:

- Automatically adapts to network changes
- Suitable for large networks

Disadvantages:

- More complex
- Uses additional bandwidth and resources

3.2.3 3. Default Routing

Default routing is used when the destination network is unknown. Packets are forwarded to a default route.

It is commonly used in small networks and internet connections.

4. Network Topology

The network consists of:

- Two routers
- Two switches
- Four PCs
- One serial connection between routers

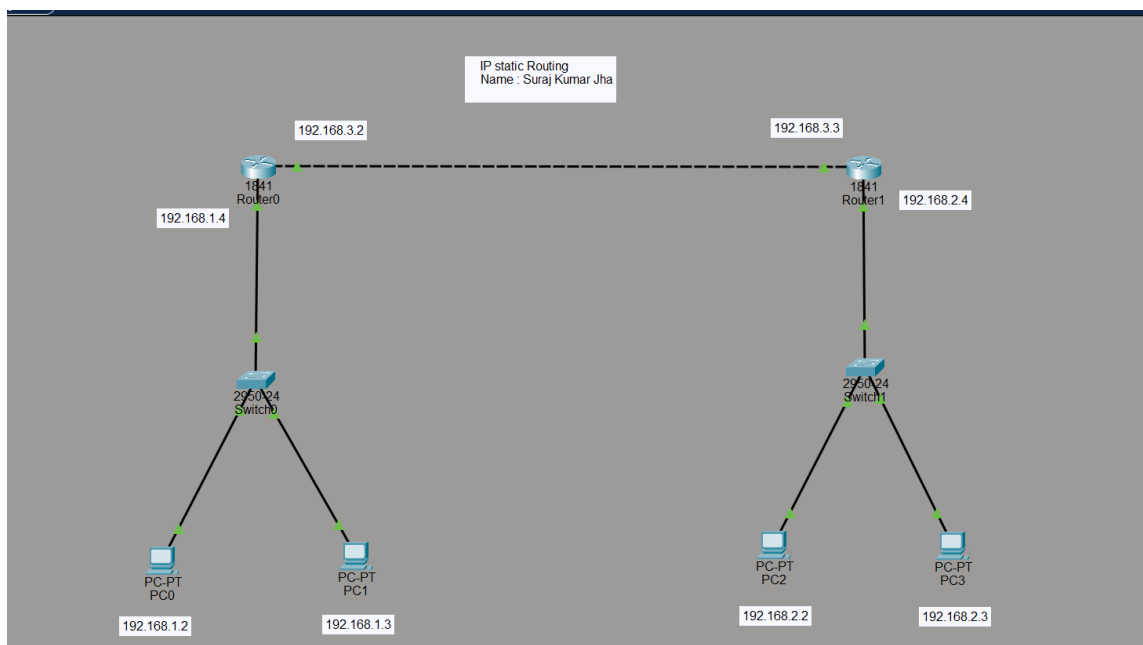


Figure 1: Static Routing Network Topology

5. IP Addressing Table

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet	192.168.1.2	255.255.255.0
PC1	FastEthernet	192.168.1.3	255.255.255.0
Router0	Fa0/0	192.168.1.4	255.255.255.0
Router0	Se0/1/0	192.168.3.2	255.255.255.0
Router1	Se0/1/0	192.168.3.3	255.255.255.0
Router1	Fa0/0	192.168.2.4	255.255.255.0
PC2	FastEthernet	192.168.2.2	255.255.255.0
PC3	FastEthernet	192.168.2.3	255.255.255.0

6. Procedure

6.1 Step 1: Create the Topology

- Open Cisco Packet Tracer.
- Add two routers, two switches, and four PCs.
- Connect PCs to switches using copper straight-through cables.
- Connect switches to routers.
- Connect routers using serial DCE cable.

6.2 Step 2: Configure PC IP Addresses

PC0

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.4

PC1

- IP Address: 192.168.1.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.4

PC2

- IP Address: 192.168.2.2

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.4

PC3

- IP Address: 192.168.2.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.4

6.3 Step 3: Configure Router0

```
enable
configure terminal

interface fa0/0
ip address 192.168.1.4 255.255.255.0
no shutdown
exit

interface se0/1/0
ip address 192.168.3.2 255.255.255.0
clock rate 64000
no shutdown
exit

ip route 192.168.2.0 255.255.255.0 192.168.3.3
```

6.4 Step 4: Configure Router1

```
enable
configure terminal

interface fa0/0
ip address 192.168.2.4 255.255.255.0
no shutdown
exit

interface se0/1/0
ip address 192.168.3.3 255.255.255.0
no shutdown
exit

ip route 192.168.1.0 255.255.255.0 192.168.3.2
```

7. Connectivity Test

Connectivity between networks was tested using the ping command.

```
ping 192.168.2.2
```

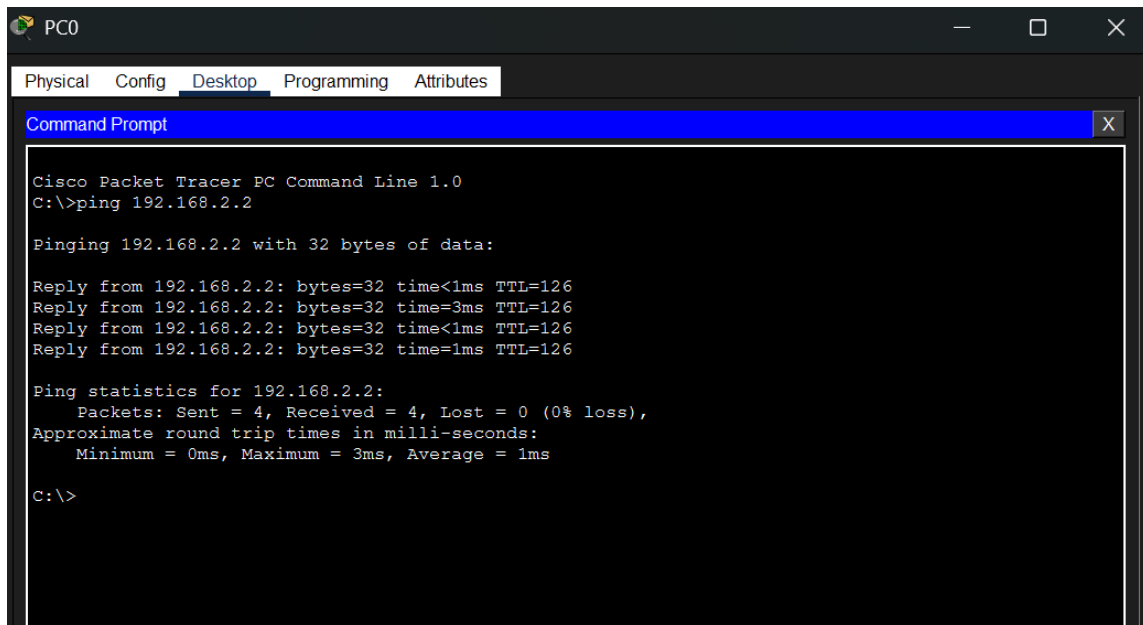


Figure 2: Connection between PC0 and PC2

Successful replies confirmed proper routing and communication between the two LAN.

A screenshot of the Cisco Packet Tracer Scenario 0 window. It shows a table of connections with columns: Fire, Last Status, Source, Destination, Type, Color, Time(sec), Periodic, Num, Edit, and Delete. The table contains three rows of successful connections.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	2	(edit)	(delete)

Figure 3: Connection result between two n/ws

8. Result

Static routing was successfully configured between two LAN networks using Cisco Packet Tracer. All devices were able to communicate with each other successfully.

9. Conclusion

This experiment demonstrated the implementation of static routing using Cisco Packet Tracer. The lab helped in understanding routing concepts, router configuration, IP addressing, and inter-network communication. Successful ping responses verified that routing was functioning correctly.